

PAP, RADs, HMMVs

Three development pathways and the current landscape

Brian W. Locke

01

**Home mechanical
ventilation**

02

**Sleep apnea and
PAP**

03

**Respiratory assist
devices**

04

**Current uses and
coverage**

History-first revision • September 19, 2026

Poliomyelitis and respiratory paralysis

1920S-1950S

- Respiratory muscle paralysis
- Negative-pressure tank ventilation
- 1952 Copenhagen: tracheotomy and prolonged positive-pressure support

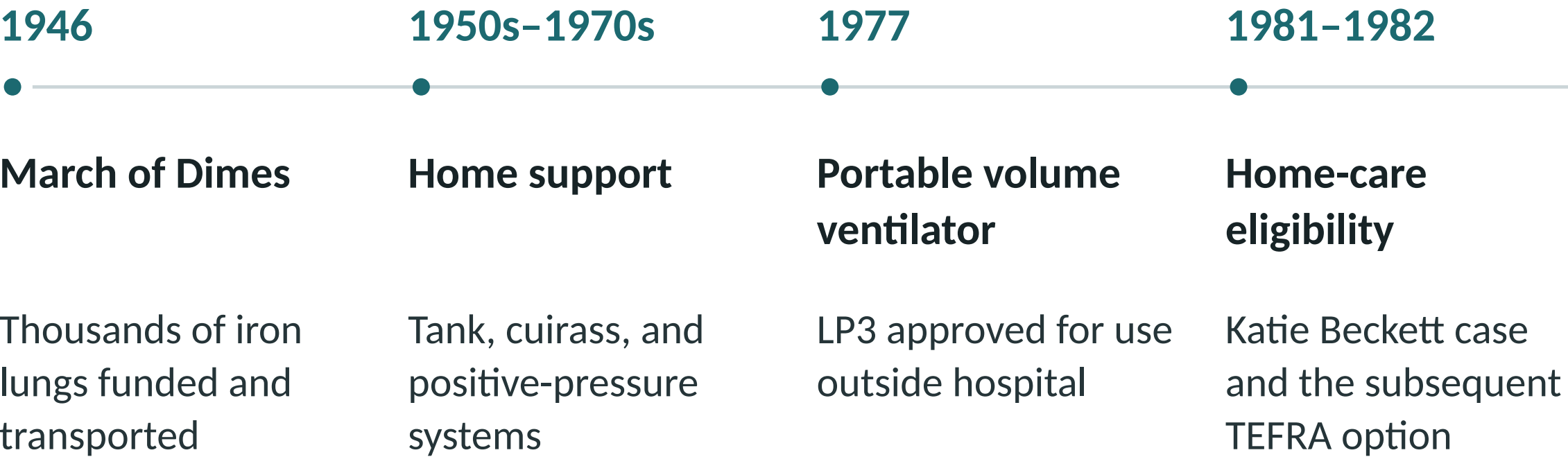
The Autumn Ghost • Hannah Wunsch, 2023



Illustrative iron lung. Date and location unspecified. Hind et al., 2017, Fig. 1.

Long-term ventilation outside hospital

Selected milestones • not to scale



Equipment portability • trained caregivers • continuing support • financing

Home ventilators: design and payment context

Clinical and engineering context

Sustained ventilatory assistance

Some patients dependent on uninterrupted support

Power, alarms, circuits, and servicing

Medicare equipment category

Frequent and substantial servicing

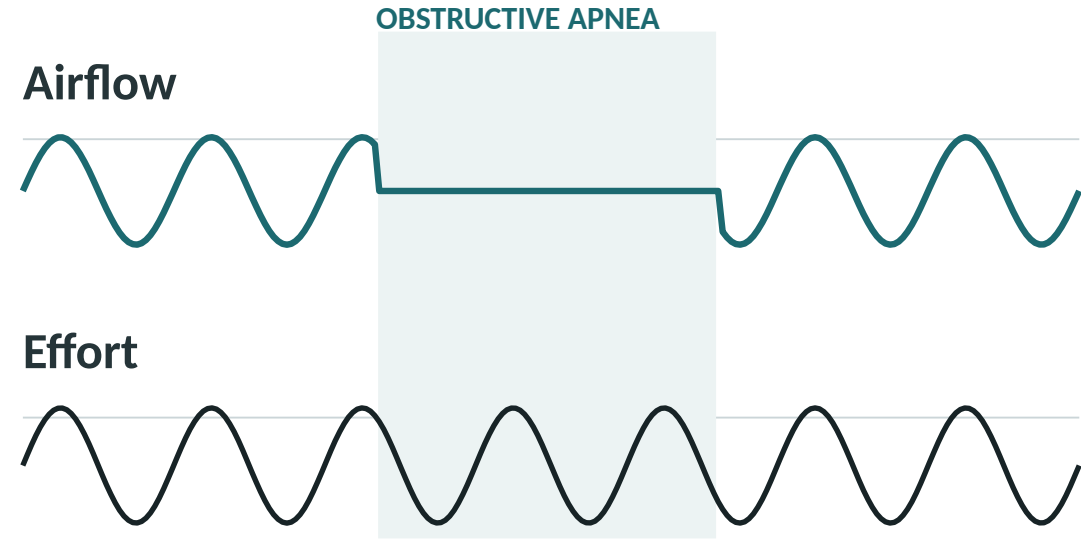
Continuing rental while medically necessary

Equipment, supplies, maintenance, and repairs

The payment category concerns equipment servicing, not a guaranteed schedule of RT visits.

Sleep apnea: the original clinical problem

- Recurrent upper-airway obstruction during sleep
- Respiratory effort persists during obstructive apnea
- Sleep fragmentation and intermittent hypoxemia



Airflow stops; effort continues.

Conceptual tracings; not patient data.

An airway-patency disorder, not necessarily daytime ventilatory failure.

Nasal CPAP, 1981

5

patients with severe
OSA

Sullivan et al. • Lancet 1981

- Pneumatic splinting of the upper airway
- No separate inspiratory pressure-support level
- 1984 report: 35 patients on home treatment for 3–30 months

4.5–10

cm H₂O in the original
report

Continuous airway pressure

No independent IPAP–EPAP difference

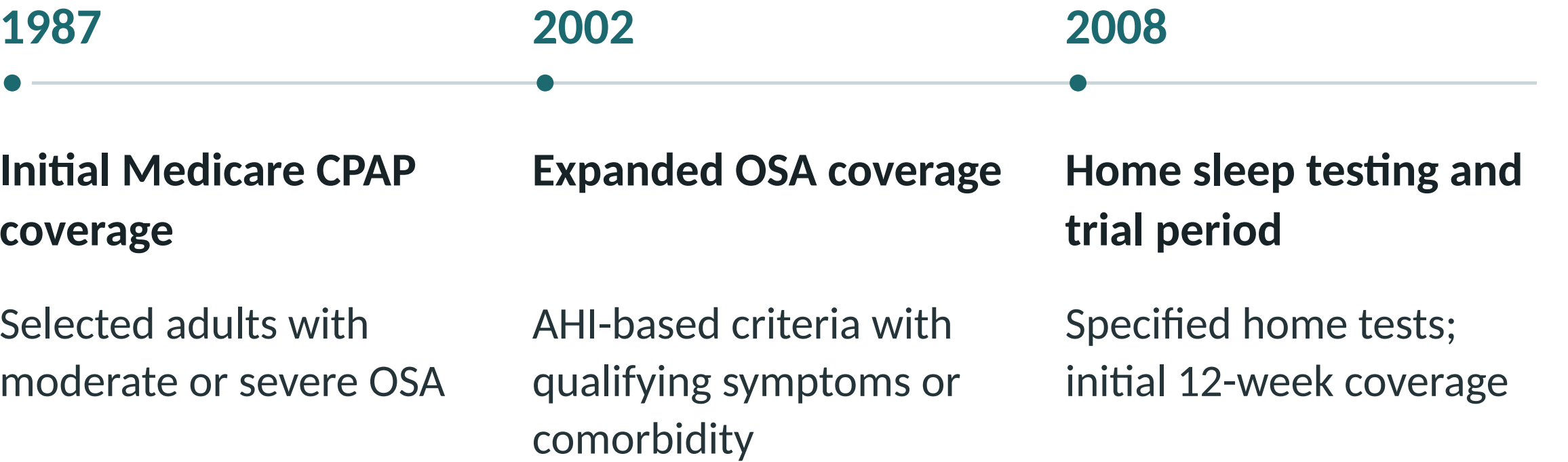
Time

A flow generator, nasal interface, and continuous positive pressure.



PAP: a sleep-apnea coverage pathway

Selected milestones · not to scale



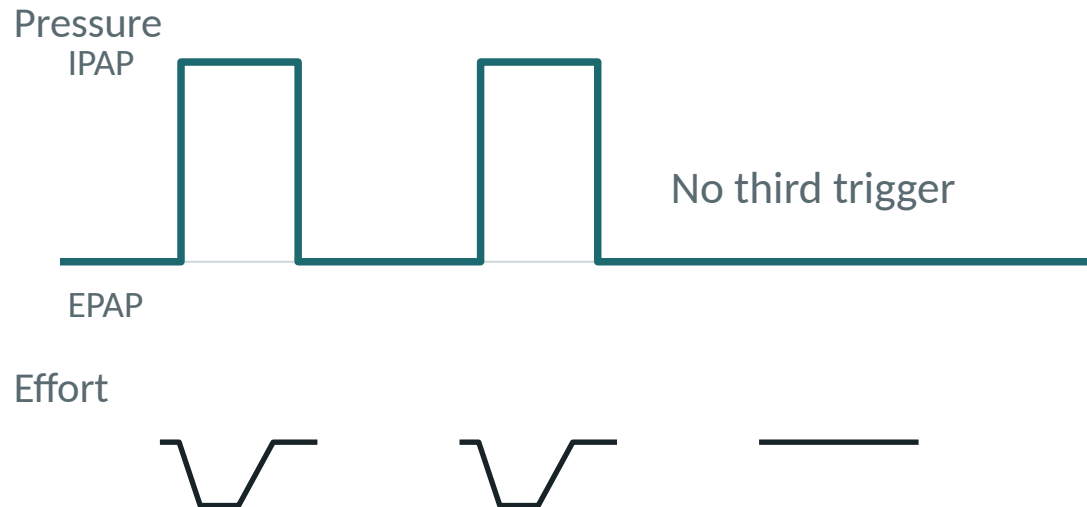
Current pathway: documented OSA, treatment benefit, and use; not daytime PaCO₂.

Bilevel positive airway pressure, 1990

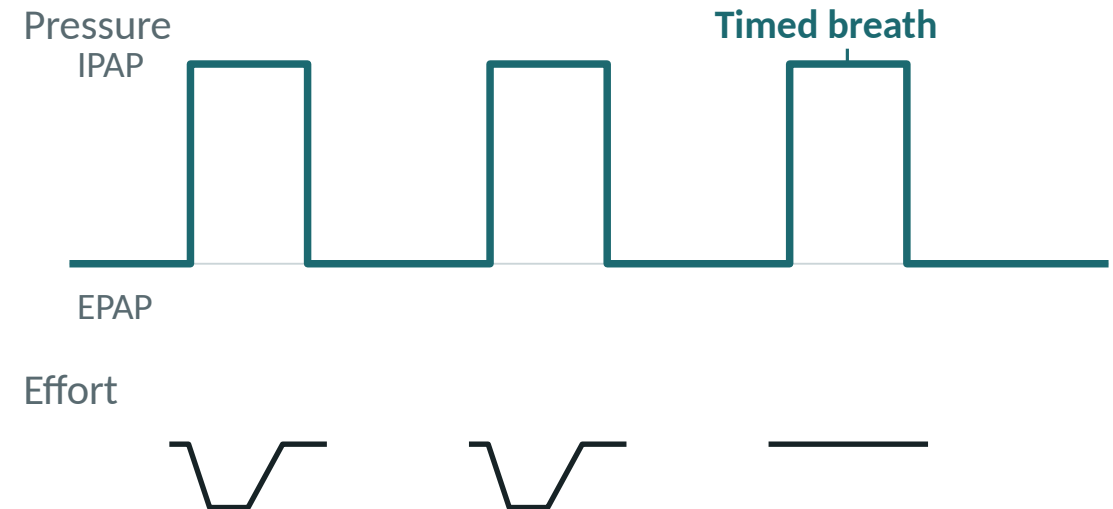
Independent inspiratory and expiratory pressures

$$PS = IPAP - EPAP$$

BILEVEL S



BILEVEL S/T



Schematic timing and amplitudes

Early bilevel technology arose in OSA; ventilatory-assistance applications expanded afterward.

NIV beyond upper-airway obstruction

Neuromuscular disease

Reduced respiratory muscle capacity

Chest-wall disorders

Increased elastic load and restricted expansion

COPD and obesity

Mechanical loading, airway disease, and drive

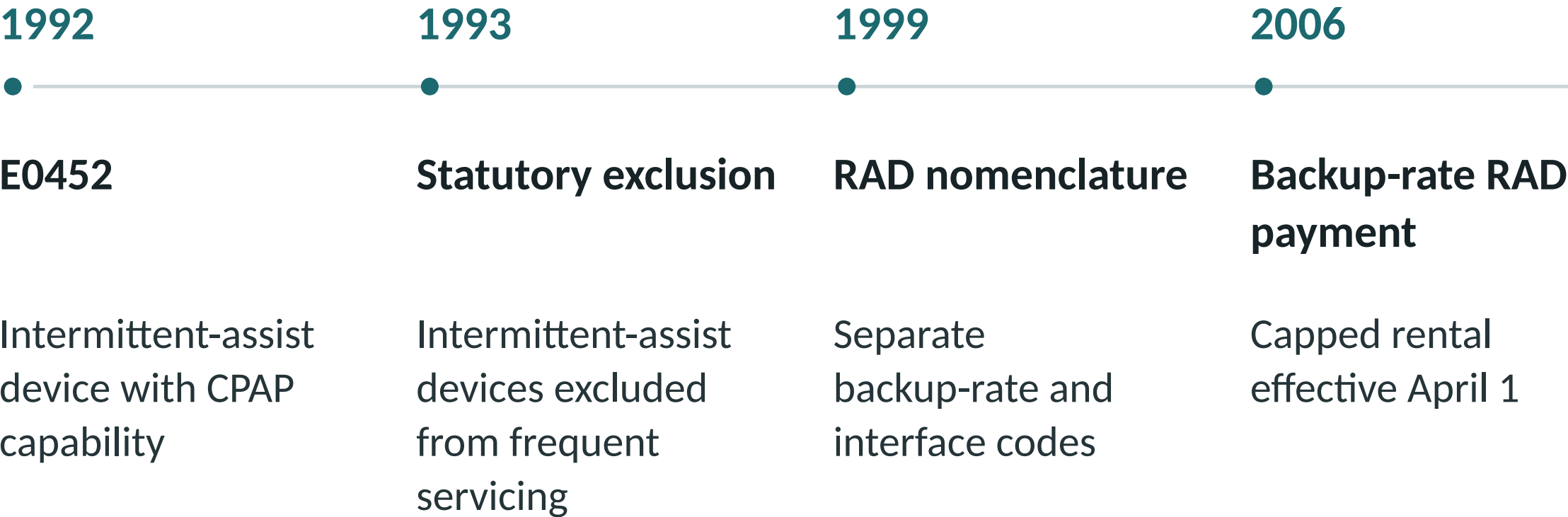
Increasing respiratory load

Reduced capacity or drive

Sleep-related hypoventilation can precede awake hypercapnia.

The equipment-servicing split, 1992–2006

Selected milestones • not to scale



Manufacturer submission: part-day use, usually during sleep, with little routine maintenance.

Diagnosis-specific RAD coverage

**Restrictive thoracic /
neuromuscular disorders**

**COPD-related
respiratory failure**

**Central / complex
sleep apnea**

**Hypoventilation
syndromes**

Coverage policies developed around diagnoses, gas exchange, testing, and device features.

Overlapping modes, different platforms

Equipment	Pressure / support	Common code
CPAP / APAP	Single-level airway pressure	E0601
Bilevel RAD	IPAP and EPAP; no backup rate	E0470
Backup-rate RAD	Bilevel with timed support	E0471
Noninvasive HMV	Ventilator-class platform	E0466
Invasive HMV	Ventilator via invasive interface	E0465

Mode availability overlaps; alarms, power, circuits, and control features remain model-specific.

Sleep-related and multifactorial hypoventilation

- Reduced compensatory drive and muscle contribution in sleep
- CO₂ accumulation during obstructive events
- Incomplete inter-event ventilation in susceptible patients



Conceptual course; not patient data.

Nocturnal impairment, daytime hypercapnia, and a single causal diagnosis are not equivalent.



Current uses: OSA, OHS, and central apnea

Clinical group	Guideline-supported approach
Routine adult OSA	CPAP or APAP
Stable OHS + severe OSA	CPAP as initial therapy
Hospitalized respiratory failure with suspected OHS	Discharge NIV pending evaluation
Central sleep apnea	Etiology-specific CPAP, backup-rate bilevel, or ASV

These are clinical guideline statements, not automatic equipment coverage.

COPD: trial populations and outcomes

Trial	Population / timing	Main result
Köhnlein 2014	Stable severe hypercapnic COPD	1-year mortality 12% vs 33%
RESCUE 2014	Hypercapnia >48 h after acute support stopped	No primary readmission / death benefit
HOT-HMV 2017	Persistent hypercapnia 2–4 weeks after acidosis	Readmission / death HR 0.49 (0.31–0.77)

Treatment strategies and selected populations, not E0471-versus-E0466 comparisons.

Nocturnal and mixed-disease evidence

Ward 2005

26 randomized
NMD / chest-wall disease
Nocturnal hypercapnia; normal
daytime CO₂

Zheng 2022

32 patients
Obesity + obstructive airways disease
Chronic hypercapnia

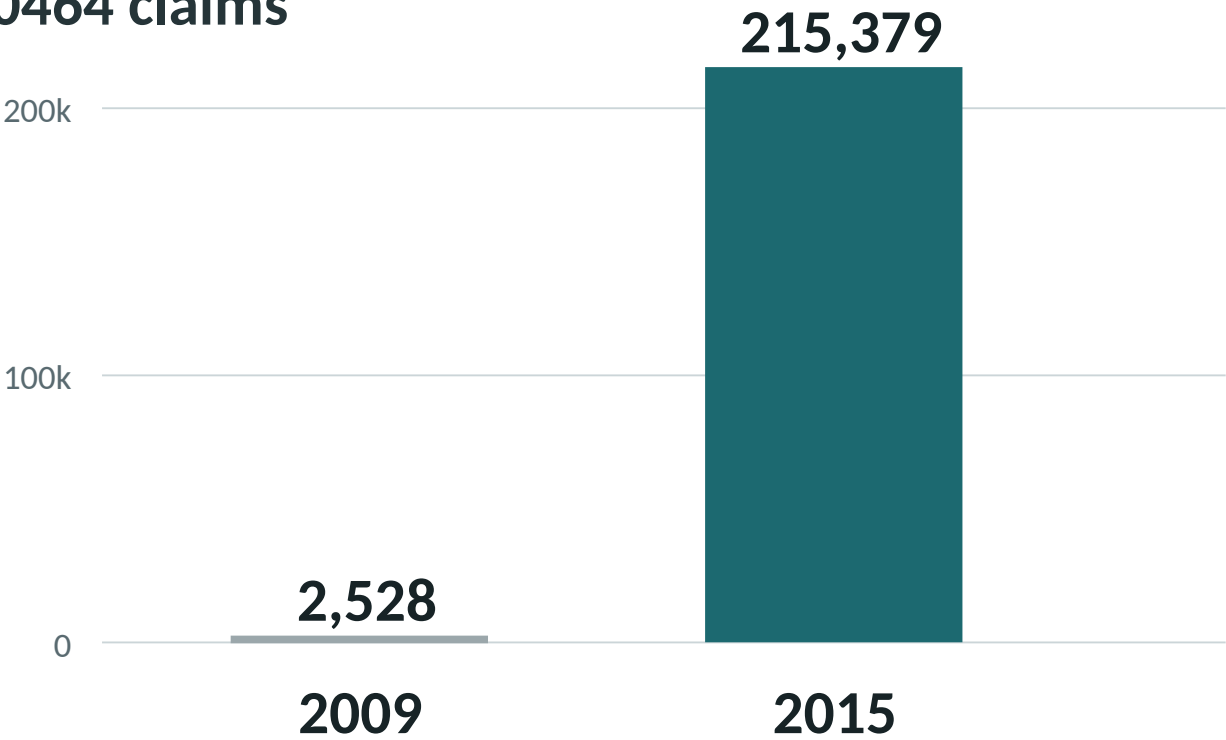
Improved nocturnal gas exchange

Bilevel-S favored for 3-month CO₂

Neither study establishes a universal treatment rule for isolated nocturnal or multifactorial hypoventilation.

Noninvasive ventilator claims, 2009–2015

E0464 claims



85×

E0464 claims

\$3.8M → \$340M

Medicare + beneficiary expenditures

415 → 32,848

Beneficiaries with claims

Historical Medicare claims analysis. Claims, beneficiaries, and clinical overuse are different measures.

Access barriers and inappropriate billing

OIG findings, 2016

\$25M with indicators of inappropriate billing

Examples: OSA treatment and overlapping device claims

No population-wide clinical overuse rate

ONMAP account, 2021

Detailed RAD qualification barriers

Reports of easier access to some ventilators

Concern about ongoing clinical support

Reported access barriers and billing concerns coexist; claims growth alone does not establish causation.



Medicare coverage terminology

Term	Role
CMS	Centers for Medicare & Medicaid Services
NCD	National Coverage Determination: national coverage policy
MAC / LCD	Medicare contractor / contractor-level coverage policy
DME supplier	Company furnishing equipment and related services
HCPCS	Equipment or service billing code

A DME supplier provides equipment. A DME MAC administers Medicare claims.

COPD national coverage, 2025

240.9

National Coverage Determination

Chronic respiratory failure
consequent to COPD

- Effective June 9, 2025
- RAD and HMO criteria in one national document
- Separate standard and hospital-discharge routes
- Non-COPD pathways remain separate

ONMAP recommendations, national coverage, and claims implementation are distinct documents.

NCD 240.9: RAD pathways

Selected entry criteria; detailed conditions in A03

Standard backup-rate route

Awake arterial PaCO₂ ≥52 mmHg

Sleep apnea not the predominant cause

Stable COPD or persistent hypercapnia ≥2 weeks after hospitalization

Immediate discharge route

Acute-on-chronic respiratory failure due to COPD

RAD or ventilator required within prior 24 hours

Clinician-assessed risk of rapid deterioration

Initial coverage: 6 months. Backup-rate therapy: IPAP ≥15 cm H₂O and rate ≥14/min by month 6.

NCD 240.9: HMV pathways

Volume-targeted mode + awake PaCO₂ ≥52 mmHg + sleep apnea not predominant

At least one additional characteristic

- 1** FiO₂ ≥36% or oxygen ≥4 L/min nasally
- 2** Ventilatory support >8 hours/day
- 3** Alarm and battery need with life-threatening interruption risk
- 4** Expected RAD inadequacy because needs exceed its capabilities

A separate discharge route requires recent ventilator use and needs beyond a RAD.

Continued coverage and documentation

COPD pathway	By 6 months	Months 7–12
RAD	Usage + one listed clinical outcome	Usage in each paid rental month
HMV	Usage criteria	Usage in each paid rental month

Usage: ≥4 hours per 24 hours on ≥70% of days

At first evaluation: qualifying 30-day period. Later: each paid rental month.

Forthcoming October 28, 2026: E0466 / E0467 / E0468 face-to-face and prior written-order requirements.

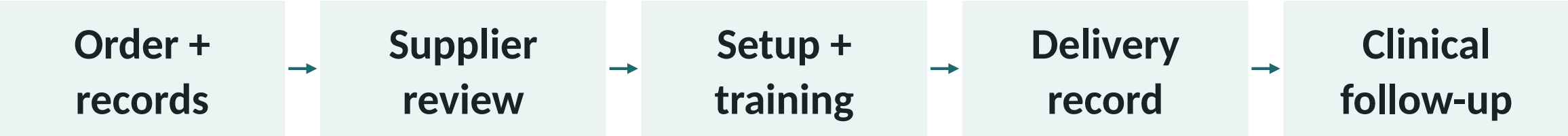


Non-COPD coverage remains diagnosis-specific

Indication	Principal policy	Distinctive requirement
OSA	NCD 240.4 / L33718	Qualifying sleep study
NMD / thoracic restriction	L33800 for RAD	Gas exchange OR NMD strength / FVC criteria
Hypoventilation syndrome	L33800	Awake PaCO ₂ + spirometry + additional findings
CSA / complex apnea	L33800	Attended PSG and treatment response

Ventilator use for NMD / thoracic restriction: NCD 280.1 and supporting justification.

DME supply and clinical follow-up



Clinical team Diagnosis, treatment parameters, response, reassessment

DME supplier Documentation intake, equipment, setup, servicing, billing

Payer / contractor Coverage rules, claims adjudication, review

Authorization, equipment delivery, and effective treatment are different events.

The contemporary landscape

PAP pathway

Sleep-apnea diagnosis
Airway-patency
treatment

RAD pathway

Bilevel assistance
Diagnosis-specific
eligibility

HMV pathway

Ventilator capabilities
Extended support
needs

Shared modes and patient populations

Overlapping physiology and modes • distinct equipment and payment rules • incomplete evidence for mixed disease

Historical chronology and source status

Milestone	Source status
1952: Copenhagen polio epidemic	Contemporary report; later historical reviews
1977: portable LP3	King review; historical FDA reference
1981: nasal CPAP	Primary clinical report
1993 / 2006: payment classification	Statute described in contemporaneous final rule
Early polio coverage intent	Uncited retrospective account in Gusman–Wolfe
Technology history, coverage history, and proposed policy motives have different evidence bases.	

Modes and control strategies

Mode or strategy	Defining feature
CPAP / APAP	One pressure level at a time; APAP adjusts that level
Bilevel S	Patient-triggered inspiratory assistance
Bilevel S/T	Spontaneous breaths plus timed backup
Volume-assured pressure support	Pressure support adjusts toward a ventilation target
Adaptive servo-ventilation	Dynamic support for selected central breathing instability
Mode names do not establish platform capabilities or a payer category.	

COPD RAD: initial coverage detail

Standard initiation

Awake PaCO₂ ≥52 mmHg, prescribed FiO₂

Sleep apnea not predominant

Backup RAD: stability or persistent post-exacerbation hypercapnia

No-backup RAD: intolerance or medical inappropriateness

Immediate discharge

Acute-on-chronic respiratory failure due to COPD

RAD or ventilator needed within prior 24 hours

Clinician-assessed risk after discharge

Separate 6-month initial coverage route

Backup-rate high-intensity minimum by 6 months: IPAP ≥15 cm H₂O and rate ≥14/min.

COPD H MV: initial coverage detail

Standard route

Volume-targeted mode

Awake PaCO₂ ≥52; sleep apnea not predominant

At least one additional need

Oxygen, duration, safety, or RAD-capability criterion

Discharge route

Acute-on-chronic COPD respiratory failure

Needs exceeded RAD capability

Ventilator required within prior 24 hours

Clinician-assessed risk after discharge

Standard and discharge pathways have different entry wording; both specify volume-targeted H MV.

Continuing coverage: three different clocks

Pathway	Evaluation and use requirements
OSA PAP	Day 31–91 reassessment; ≥ 4 h on $\geq 70\%$ of nights in a 30-day period
Non-COPD RAD	Reevaluation no sooner than day 61; benefit and average ≥ 4 h/day
COPD RAD	By 6 months: use threshold plus ≥ 1 CO ₂ or symptom outcome
COPD HMV	By 6 months: use threshold; no parallel outcome list
COPD RAD / HMV	Months 7–12 and later: monthly continuing-use requirements

The familiar OSA adherence window is not a universal rule for every home-ventilation pathway.

OSA PAP: diagnostic and coverage details

Initial evidence

Clinical evaluation and qualifying sleep test

AHI/RDI ≥ 15 , or 5–14 with specified findings

Medicare hypopnea definition: $\geq 4\%$ desaturation

Initial national coverage period: 12 weeks

Device and continuation

E0601: CPAP/APAP

E0470: E0601 tried and ineffective

E0471: not covered for primary OSA

Clinical benefit plus defined adherence window

PAP in the OSA policy includes E0601 and qualifying E0470 equipment.

Non-COPD RAD: restrictive and hypoventilation routes

Restrictive thoracic / NMD

Appropriate diagnosis; COPD not significant
PaCO₂ ≥45 OR qualifying nocturnal desaturation
NMD alternative: MIP <60 or FVC <50%

Either E0470 or E0471

Hypoventilation syndrome

Awake PaCO₂ ≥45 and FEV₁/FVC ≥70%

Additional sleep/waking gas or nonobstructive desaturation finding
E0471 has a covered-E0470 step-up structure
Symptoms and full documentation apply

The NMD muscle-strength and FVC alternatives do not require an elevated daytime PaCO₂.

Central apnea: coverage and clinical guidance

LCD coverage

Attended facility polysomnography

AHI and CAHI ≥ 5 ; central events $> 50\%$

CSA definition excludes hypoventilation

Qualifying response on prescribed settings

2025 AASM guidance

Etiology-specific CPAP, backup bilevel, or ASV

Bilevel without backup suggested against

HFrEF ASV: experienced-center monitoring

Clinical recommendations differ from code eligibility

Either E0470 or E0471 may satisfy the LCD; that permission is not a recommendation for every mode.

COPD trial comparison

Trial	Population	Principal result
Köhnlein 2014	Stable severe hypercapnic COPD	1-year mortality: 12% vs 33%
RESCUE 2014	Hypercapnia >48 h after acute support	No primary readmission/death benefit
HOT-HMV 2017	Persistent hypercapnia 2–4 weeks after acidosis resolved	HR 0.49 (95% CI 0.31–0.77)

Selection, timing, comparator, and ventilation strategy differed across trials.

Nocturnal and overlap populations

Study	Direct evidence	Limits
Ward 2005 26 randomized	NMD/chest-wall disease; nocturnal hypercapnia; daytime normocapnia	Small, mixed-age selected population
Zheng 2022 32 randomized	Obesity + airway obstruction + chronic hypercapnia; BPAP-S vs CPAP	3-month CO ₂ endpoint; not an HMV comparison
An observed nocturnal abnormality does not have one validated treatment threshold across all etiologies.		

Measures of home-treatment performance

Domain	Examples	Interpretation
Use	Hours; days; interruptions	Exposure to treatment
Technical support	Leak; triggering; residual events	Device-specific estimates
Gas exchange	Awake ABG; nocturnal CO ₂ / SpO ₂	Timing and measurement method matter
Patient outcomes	Symptoms; sleep; function; admissions	Not interchangeable with usage
Association between CO ₂ reduction and outcome is not proof of a universal causal treatment target.		

Documentation and source index

Topic	Source
COPD RAD / HMO	NCD 240.9 and CMS FAQ
OSA PAP	NCD 240.4; LCD L33718
Non-COPD RAD	LCD L33800
Other ventilator indications	NCD 280.1; contractor coding guidance
F2F / prior written order	CMS order requirements; Oct 28, 2026 expansion
Utah Medicaid	State criteria and applicable plan process
Policy reference date: September 19, 2026. E0466/E0467/E0468 F2F/WOPD expansion is forthcoming.	